

Point-by-Point Response to Technical Check

npj Artificial Intelligence

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Title: Financial Time-Series Generation: A Systematic Survey of Taxonomy, Evaluation, Datasets, and

Authors: Yingxiao Zhang, Jiaxin Duan, Yue Zou, Ke Feng, Jing Xiao

Assistant Editor: Dr. Monika Samra

September 5, 2026

Dear Dr. Monika Samra and Editorial Team of *npj Artificial Intelligence*,

We sincerely appreciate the editorial team's diligent technical review of our manuscript. We have carefully addressed every point raised in the Technical Check letter. The manuscript has been comprehensively restructured into the required **IRDM** format, unnumbered section headings have been adopted, the abstract has been condensed to a single paragraph of 179 words, all declarations have been fully integrated, references have been converted to the sequential Nature numbering style, figure legends have been thoroughly enhanced, and a completed PRISMA 2020 checklist has been prepared as Supplementary Information.

Below is our point-by-point response detailing the specific revisions made to address each comment.

Point 1: Author Affiliations and Formatting

Editor's Comment: *"Please ensure an affiliation, including city and country information, or a URL, is provided on the title page for all authors, including for the corresponding author(s). It should be made clear which affiliation belongs to each author e.g. by using a numbered list or using superscript numbering. Please ensure that all author affiliations appear in the correct sequential order according to their position in the author list. Affiliation 1 must be associated with the first author."*

Response: We have thoroughly updated the title page of the manuscript. All author affiliations are now presented using superscript numbers (^{1,2,3,4}) appearing in strict sequential order matching their appearance in the author list. Affiliation 1 is associated with the first author, Yingxiao Zhang. Each affiliation explicitly includes the department/school, institution, city, and country:

1. ¹School of Software and Microelectronics, Peking University, Beijing, China.
2. ²Pingan Bank Co., Ltd., Shenzhen, China.
3. ³China Electronics Cloud Technology Co., Ltd., Beijing, China.

4. ⁴Pingan Group Co., Ltd., Shenzhen, China.

Both corresponding authors (Ke Feng and Jing Xiao) are clearly designated with asterisks (*) and their institutional email addresses are provided on the title page.

Point 2: Abstract Formatting and Word Count

Editor’s Comment: *“The journal does not permit a structured abstract. The abstract must be a single paragraph with no headings (e.g. Methods, Results, Conclusions) or paragraphs, should be around 150-200 words, and written in the English language. Please also ensure you also update the Abstract in the submission system.”*

Response: We have restructured and condensed the abstract into a **single cohesive paragraph of exactly 179 words** without any subheadings, bold headers, or paragraph breaks. The revised abstract succinctly synthesizes the motivation, two-level taxonomy, evaluation protocols, benchmark catalogs, and open challenges. We confirm that this identical 179-word text has also been synchronized in the online SNAPP submission system.

Point 3: Removal of Section Numbers and Cross-References

Editor’s Comment: *“Please note that section numbers will not be retained during typesetting of the manuscript. Section numbers must be removed from the headings, and reference to section numbers in the main text must be changed to the section headings.”*

Response: We have configured ‘\setcounter{secnumdepth}{0}’ in the LaTeX preamble to remove all numerical identifiers from section, subsection, and subsubsection headings across the entire manuscript. Furthermore, we conducted a rigorous full-text audit and replaced all textual and parenthetical section number references (e.g., “Section 2”, “(§??)”, “Sections 3–6”, “§8”) with their corresponding explicit section titles (e.g., “the Problem Formulation and Taxonomy subsection”, “the Evaluation Protocols subsection”, “the Applications subsection”, “the Methods section”, etc.).

Point 4: Unstructured Introduction

Editor’s Comment: *“We do not allow a structured Introduction with subheadings. It can only be divided into paragraphs.”*

Response: All four previous subheadings in the Introduction (“Background and Motivation”, “Why Existing Surveys are Insufficient”, “Scope and Contributions”, and “Organization of the Paper”) have been removed. The Introduction now progresses smoothly through continuous, coherent paragraphs. The final paragraph has also been updated to explicitly outline the IRDM workflow of the manuscript.

Point 5: IRDM Structural Organization

Editor’s Comment: *“Format of Research Article: IRDM. Please arrange your manuscript according to the below order: Abstract, Introduction, Results (subheadings compulsory), Discussion (no subheadings permitted, incorporate Limitations/Conclusions), Methods (subheadings compulsory), Declaration statements, References list, Figures, Tables and associated legends...”*

Response: The entire manuscript has been reorganized to strictly adhere to the IRDM structure:

- **Abstract:** Single paragraph, 179 words, unnumbered, no subheadings.
 - **Introduction:** Unstructured, no subheadings.
 - **Results:** Features compulsory thematic subheadings: (1) Problem Formulation and A Two-Level Taxonomy; (2) Financial Time-Series Extrapolation; (3) Financial Time-Series Imputation; (4) Financial Time-Series Synthesis; (5) Applications; (6) Evaluation Protocols; (7) Data Resources and Benchmarks.
 - **Discussion:** Contains no subheadings (per editorial guidelines); organizes open challenges, limitations, and concluding outlook via bold paragraph lead-ins.
 - **Methods:** Positioned directly after Discussion with compulsory subheadings: (1) Literature Retrieval Strategy; (2) Inclusion and Exclusion Criteria; (3) Data Extraction and Coding Protocol; (4) Review Dimensions and Taxonomy Synthesis.
 - **Declarations:** Follows Methods and precedes References.
 - **References:** Nature-compliant sequential numbered bibliography.
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Point 6 & 7: Renaming Survey Sections and Incorporating Conclusions

Editor’s Comment: *“We do not permit ‘Methodology of the Survey, Problem Formulation and A Two-Level Taxonomy, Financial Time-Series Extrapolation, Financial Time-Series Imputation, Applications, Financial Time-Series Synthesis, Evaluation Protocols, Data Resources and Benchmarks, Open Challenges and Future Directions’ sections. Please rename it to ‘Introduction/Methods/Discussion’. We do not permit a Conclusions section. Please rename it to ‘Discussion’ or incorporate the text into the ‘Discussion’ section, as applicable.”*

Response: In accordance with this directive:

1. The previous “Methodology of the Survey” section has been renamed to “Methods” and placed after Discussion.
2. The taxonomic and synthesis bodies (Problem Formulation, Extrapolation, Imputation, Synthesis, Applications, Evaluation Protocols, Data Resources) have been nested under the “Results” section with compulsory subheadings.

3. The “Open Challenges and Future Directions” content and the former “Conclusions” section have been fully integrated into the unified “Discussion” section under the thematic lead-in “Limitations, practical implications, and concluding outlook”, without any subheadings.
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Point 8: Acknowledgements and Funding Statements

Editor’s Comment: *“Please supply an Acknowledgements section before the References... Please supply a Funding section after the Acknowledgements section. Please include all funding information, including the role of the funder(s)... If no funding was granted for the study, please also state this in the Funding Statement.”*

Response: Both sections have been added before the References, as required:

- **Acknowledgements:** The section is placed immediately before the References and reads: “The authors would like to thank the colleagues and members of the research group for constructive discussions that contributed to refining the scope and taxonomy of this survey.” This single-sentence statement is kept deliberately brief, is free of any thanks to editors or reviewers, contains no effusive praise or dedicatory language, and is consistent with the journal’s guidance for the Acknowledgements section. According to the guidance, if no acknowledgement content exists, “Not applicable” should be stated under this heading; in our case, a genuine acknowledgement of colleagues exists, so the sentence above is retained rather than “Not applicable.”
- **Funding:** As no external grants were received, the official standard non-funding declaration has been inserted: “The authors declare that no funds, grants, or other financial support were received during the preparation of this manuscript.”

Both statements have been verified verbatim in the online SNAPP submission portal under the corresponding declarations fields.

Point 9: Competing Interests Statement

Editor’s Comment: *“Authors must disclose and specify any competing interest, using the following statement as applicable. No Competing Interests: ‘The authors declare no competing financial or non-financial interests.’... The competing interests statement must be included both in the manuscript and in the submission system.”*

Response: We have included the exact required sentence under the mandatory heading in the manuscript:

Competing Interests: “The authors declare no competing financial or non-financial interests.”

This identical statement has also been confirmed in the SNAPP submission portal.

Point 10: Reference List in Nature Format

Editor’s Comment: *“Please ensure the references are in the standard Nature format. References should be numbered sequentially as they appear in the text, methods, tables, and figure legends using a 1, 2, 3, format...”*

Response: We have fully migrated the citation and bibliography pipeline to the official Nature sequential numerical format using `naturemag.bst`. All hidden pre-citation macros have been eliminated, ensuring that references are numbered strictly in the order of their first appearance in the visible text: superscript 1 (Zhao et al., 2025) and superscript 2 (Jiang et al., 2025) on page 1 of the Introduction, followed sequentially by 3, 4, 5... through 130. The reference list entries appear as 1., 2., 3.... 130. in Nature’s sequence format (author list, paper title, journal name in italics, volume number in bold, initial–final page numbers, and publication year).

Point 11: Individual Figure Files, Relocation after References, and Extended Legends

Editor’s Comment: *“Please provide figure files individually in the system... Figure titles and legends must be removed from the image file... Add the legends in the main manuscript file after references. Begin each figure legend with a concise sentence summarizing the content of the figure; this will serve as the figure title. This should be followed by a detailed description of the figure in the following sentences, up to 350 words, to serve as the legend body. Define all abbreviations, symbols, and color codes used in the figure. Do not use graphical symbols directly, describe them in words... Present multi-panel figures on a single page, labeled a), b), c), etc...”*

Response: All requirements regarding figures, tables, and legends have been systematically addressed:

1. **Repositioned after References:** Per the journal’s IRDM layout instructions, all figure environments and tables have been moved from the main body and placed into dedicated sections after the References list (**Figure Legends and Figures and Tables**).
2. **Figure Numbering and Callout Order:** All figure callouts in the text now appear in strict chronological order from Figure 1 through Figure 6.
3. **Extended Legends:** Each figure caption begins with a concise bold summary sentence serving as the title (e.g., **Figure 1 | Evolution of Financial Time-Series Generation techniques over the past decade.**), followed by a comprehensive description (under 350 words) defining all subpanels (**a**, **b**, **c**), line styles, and color codes in plain English words without raw graphical symbols. All acronyms (e.g., NMTucker, RegTensor, NDDP, CSDI, MarS) are explicitly defined.
4. **Clean Standalone Figure Files:** High-resolution standalone figure files (**Figure1.pdf**, **Figure2.pdf**, **Figure3.png**, **Figure4.pdf**, **Figure5.pdf**, and **Figure6.png**) have been exported into a dedicated folder (**separate_figures/**) completely stripped of embedded figure labels (e.g., “Figure 1”) and titles, formatted for individual upload to the submission system.

Point 12: Clean and Marked-Up Manuscript Files

Editor’s Comment: *“Please note that we require both clean and marked-up versions of the manuscript at this stage... provide your clean article file without any red text as a Manuscript file. Please upload the marked-up version of the manuscript to the submission system as a Related manuscript file.”*

Response: We have prepared both required PDF documents:

- **Clean version (is26.pdf):** A completely clean, unannotated manuscript in strict IRDM format with Nature numbered references, unnumbered headings, and post-reference float layout, ready for direct production typesetting.
- **Marked-up version (is26_marked.pdf):** Features a prominent revision banner and a dedicated summary block detailing all technical modifications across affiliations, abstract condensation, IRDM restructuring, declaration integration, citation re-indexing, and float repositioning, submitted as a Related manuscript file.

Point 13: Data Availability and Code Availability Declarations

Editor’s Comment: *“The Data Availability information must also be uploaded to system, and the details provided there should match exactly with the statement in the manuscript.”*

Response: Both statements have been placed in the manuscript and synchronized in the submission portal:

- **Data Availability:** “No new empirical primary datasets were generated during the current study. All benchmark datasets, curated databases, and public APIs reviewed and analyzed in this survey (including FinTSB, TFB, TSGBench, CTBench, FinTSBridge, WRDS, CRSP, and FRED) are derived from published literature and publicly accessible repositories as referenced in the Data Resources and Benchmarks subsection. The literature retrieval query logs and systematic extraction coding tables supporting the findings of this review are available from the corresponding authors upon reasonable request.”
- **Code Availability:** “This work is a systematic literature survey and did not generate custom algorithmic codebases or software artifacts. All generative models, benchmark evaluation suites, and foundational model repositories discussed are publicly accessible via their respective cited publications and open-source repositories as cataloged throughout the manuscript.”

Point 14: Figure Copyright and Originality Confirmation in Cover Letter

Editor’s Comment: *“Please state in your cover letter whether all Figures are your own or taken from another source...”*

Response: We have prepared and submitted a formal **Cover Letter** (Cover_Letter.pdf) addressed to Assistant Editor Dr. Monika Samra. The cover letter explicitly certifies:

“We formally confirm that all figures (Figures 1, 2, 3, 4, 5, and 6) included in this manuscript and supplementary materials are original creations by the authors. Figure 1 was designed using LaTeX TikZ, Figure 2 was constructed using LaTeX Forest, and Figures 3–6 were plotted directly by the authors using custom scientific diagramming tools. No figure, subpanel, or graphical element has been reproduced, adapted, or taken from third-party publications or copyrighted external sources. Therefore, no external copyright permissions or licensing authorizations are required.”

Point 15: Author Contribution Initials Disambiguation

Editor’s Comment: *“Please note that Author contribution statement provided in manuscript should exactly match with that uploaded in the system.”*

Response: To prevent ambiguity in the submission system and author registry where two co-authors share identical initials (Yingxiao Zhang and Yue Zou), we have explicitly disambiguated their citations in the Author Contributions section:

Author Contributions: “Y. Zhang: Conceptualization, Methodology, Writing – Original Draft. J. Duan: Validation, Formal analysis, Writing – Review & Editing. Y. Zou: Validation, Formal analysis, Writing – Review & Editing. K. Feng: Supervision, Funding acquisition, Project administration. J. Xiao: Supervision, Funding acquisition, Project administration. All authors reviewed, discussed, and approved the final submitted version of the manuscript.”

This updated statement has been matched verbatim in the online SNAPP portal.

Point 16: Mandated Systematic Review Checklist (PRISMA 2020)

Editor’s Comment: *“Please note that systematic reviews and meta-analyses papers must adhere to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) or PRISMA-SCR for scoping reviews. It is mandatory to provide a completed PRISMA checklist (or appropriate extension) in the Supplementary Information.”*

Response: We have prepared a dedicated Supplementary Information document (Supplementary_Information) containing the complete, itemized **PRISMA 2020 Checklist** across all 27 standard reporting items, detailing the exact manuscript location and page number where each item is addressed.

We believe these comprehensive revisions fully satisfy all editorial requirements of *npj Artificial Intelligence*. Thank you again for your valuable editorial guidance.

Sincerely,

The Authors

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Peking University & Pingan Group